Universität Siegen

Naturwissenschaftlich-Technische Fakultät

Master's Thesis

Evaluating and Improving Software Development Processes in Large-Scale Projects Using LLM-Based AI Agents: A Case Study of CrypTool

Examiner Prof. Dr. Bernhard Esslinger
Prof. Dr. Roland Wismüller

Author Michael Heep

Submitted

Kurzzusammenfassung

Abstract

Contents

Kurzzusammenfassung 2

Abstract 2

1 Introduction 4
- 1.1 Motivation and Problem Statement . . . 4
- 1.2 Research Questions . . . 4
- 1.3 Limitations for this Master's Thesis . . . 4
- 1.4 Structure . . . 4

2 Background and Related work 5

3 Foundations 6
- 3.1 Large Language Models (LLMs) . . . 6
- 3.2 Agentic AI . . . 6
- 3.3 Skills and Tooling . . . 6
- 3.4 Anforderungen im CTO-Entwicklungsprojekt . . . 6

4 Developing an Agent Setup for CrypTool-Online 7

5 Implementation and Case Studies 8
- 5.1 Experimental Setup . . . 8
- 5.2 Case Study: Claude Code . . . 8
- 5.3 Case Study: Codex . . . 8
- 5.4 Case Study: OpenCode . . . 8

6 Evaluation 9

7 Conclusion and Future Work 10

1 Introduction

1.1 Motivation and Problem Statement

1.2 Research Questions

1.3 Limitations for this Master's Thesis

1.4 Structure

2 Background and Related work

3 Foundations

3.1 Large Language Models (LLMs)

3.2 Agentic AI

3.3 Skills and Tooling

3.4 Anforderungen im CTO-Entwicklungsprojekt

4 Developing an Agent Setup for CrypTool-Online

5 Implementation and Case Studies

5.1 Experimental Setup

5.2 Case Study: Claude Code

5.3 Case Study: Codex

5.4 Case Study: OpenCode

6 Evaluation

7 Conclusion and Future Work